

# Tournament Graph Distribution

Time limit: 1.00 s

Memory limit: 512 MB

A *tournament graph* is a directed graph where a single directed edge exists between every pair of nodes.

Given  $n$ , your task is to calculate for each  $k = 1 \dots n$  the number of tournament graphs that have  $n$  nodes and  $k$  strongly connected components.

## Input

The only line has an integer  $n$ : the number of nodes.

## Output

Print  $n$  lines: for each  $k = 1 \dots n$  the number of graphs modulo  $10^9 + 7$ .

## Constraints

- $1 \leq n \leq 500$

## Example

| Input | Output      |
|-------|-------------|
| 3     | 2<br>0<br>6 |